

CDS6324: Data Visualization

Trimester : (March 2025)

Assignment 1 – Airline Traffic

Individual Work

Data

The data for this assignment is provided in the **Airline_2510.twbx** file. This dataset offers an in-depth view of domestic airline routes across the United States, containing valuable insights for examining passenger travel behavior, market dynamics, and airline pricing strategies. The period covered is 1993-2024.

The dataset includes several key features relevant to domestic airline routes. These features are: the year and quarter in which the data was recorded; unique City Market IDs to identify travel markets; the names of the departure and arrival cities; and the distance between these cities measured in miles. Additionally, the dataset provides the average number of passengers traveling on each route per day, as well as the average fare paid by passengers, which can be further specified with currency information for clarity.

The Assignment

Assume you are a Data Analyst in a consulting firm and presenting the trends of airline travel to a potential investor in the aviation industry. The investor wants to launch a new **budget airline** on the market.

You are required to speculate/find out the questions that the investor may have regarding this industry. Based on these questions, you are required to build a story for the visualization to convey the key insights and trends in this industry that will help your client with business decision making.

Example questions:

- Which are the busiest airports in a particular year?
- Which routes have the highest revenue?
- Which routes have the lowest average fare?
- Are there any seasonal trends (more flights in summer)?
- Which region in the United States have more airline activities?
- Which destinations are growing in demand over time?

Building Data Visualization

Observe and study the data. **Think broadly about the data.** Think about what kind of information the data provides. How to extract meaningful insights from the data that is available? You may want to combine data from multiple columns to produce more meaningful metrics/information.

Your data visualization should take the form of a dashboard featuring multiple charts that help answer a range of interesting questions (3 to 6 worksheets are ideal for this assignment). While sample questions have been provided above, feel free to explore beyond them. Be creative and think of the best ways to present your data. Consider having multiple panels on the same worksheet. Explore different chart types (time series, bars, geospatial, etc.) and colour themes.

Submission Details

Your submission should consist of a single Tableau packaged workbooks(.twbx) comprising all relevant files that you use. Name your file as StudentID_StudentName.twbx and submit it to **eBwise**.

This is an **INDIVIDUAL** assignment.

Submission deadline: 24th April 2025, 11:59pm (Week 5)

Points: 10%

APPENDIX

The description of the fields is given below:

- **Year:** The year the data was recorded
- **quarter:** The quarter of the year
- **citymarketid_1:** Numeric ID code for the origin city market
- **citymarketid_2:** Numeric ID code for the destination city market
- **city1:** Origin city name (e.g., "Allentown/Bethlehem/Easton, PA")
- **State1:** Origin State Code
- **Statename1:** Origin State Name
- **Region1:** Origin Region Name
- **city2:** Destination city name (e.g., "Tampa, FL")
- **State2:** Destination State Code
- **Statename2:** Destination State Name
- **Region2:** Destination Region Name
- **airportid_1:** Numeric ID code for the origin airport
- **airportid_2:** Numeric ID code for the destination airport
- **airport_1:** Three-letter code for the origin airport (e.g., "ABE" for Allentown)
- **airport_2:** Three-letter code for the destination airport (e.g., "TPA" for Tampa)
- **nsmiles:** Distance between airports in miles
- **passengers:** Average number of passengers who flew this route during the given time period
- **fare:** Average fare price in dollars for this route
- **carrier_lg:** Code for the largest carrier on this route (e.g., "AA" for American Airlines)
- **large_ms:** Market share of the largest carrier (as a decimal, where 1 = 100%)
- **fare_lg:** Average fare price charged by the largest carrier
- **carrier_low:** Code for the lowest-fare carrier on this route
- **lf_ms:** Market share of the lowest-fare carrier (as a decimal)
- **fare_low:** Average fare price charged by the lowest-fare carrier